

ASSIGNMENT - 5

Q=Write a C program to read and print an Employee's Details using Structure - You have to read employee's details like name, employee id and salary using a structure and then print the entered values

```
#include <stdio.h>

int main() {

    char id[10];

    int hour;

    double value, salary;

    printf("Input the Employees ID(Max. 10 chars): ");

    scanf("%s", &id);

    printf("\nInput the working hrs: ");

    scanf("%d", &hour);

    printf("\nSalary amount/hr: ");

    scanf("%lf", &value);

    salary = value * hour;

    printf("\nEmployees ID = %s\nSalary = U$ %.2lf\n", id,salary);

    return 0;

}
```

```
Input the Employees ID(Max. 10 chars): 12345
Input the working hrs: 8
Salary amount/hr: 500000
Employees ID = 12345
Salary = U$ 4000000.00
```

Q=Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

```
#include<stdio.h>
```

```

/* Declaring Structure */

struct complex
{
    float real;
    float imaginary;
};

int main()
{
    /* Declaring structure variable using struct complex */
    struct complex cnum1, cnum2, sum;

    printf("Enter real and imaginary part of first complex number:\n");
    scanf("%f%f", &cnum1.real, &cnum1.imaginary);

    printf("Enter real and imaginary part of second complex number:\n");
    scanf("%f%f", &cnum2.real, &cnum2.imaginary);

    sum.real = cnum1.real + cnum2.real;
    sum.imaginary = cnum1.imaginary + cnum2.imaginary;

    printf("SUM = %0.2f + i %0.2f", sum.real, sum.imaginary);

    return 0;
}

```

```

Enter real and imaginary part of first complex number:
7 9
Enter real and imaginary part of second complex number:
5
4
SUM = 12.00 + i 13.00

```

Q=Write a C Program to store Information of 5 Students Using Array of Structure

```

#include<stdio.h>
struct student
{
    char name[20];
    int id;
    float marks;
};
void main()
{
    struct student s1,s2,s3;
    int dummy;
    printf("Enter the name, id, and marks of student 1 ");
    scanf("%s %d %f",s1.name,&s1.id,&s1.marks);
    scanf("%c",&dummy);
    printf("Enter the name, id, and marks of student 2 ");
    scanf("%s %d %f",s2.name,&s2.id,&s2.marks);
    scanf("%c",&dummy);
    printf("Enter the name, id, and marks of student 3 ");
    scanf("%s %d %f",s3.name,&s3.id,&s3.marks);
    scanf("%c",&dummy);
    printf("Printing the details....\n");
    printf("%s %d %f\n",s1.name,s1.id,s1.marks);
    printf("%s %d %f\n",s2.name,s2.id,s2.marks);
    printf("%s %d %f\n",s3.name,s3.id,s3.marks);
}

```

Enter the name, id, and marks of student 1 kratik 12 45
 Enter the name, id, and marks of student 2 sar 25 45
 Enter the name, id, and marks of student 3 1
 14 25
 Printing the details....
 kratik 12 45.000000
 sar 25 45.000000
 1 14 25.000000

Q=Write a program in C to create and store information in a text file

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    char str[1000];
    FILE *fptr;
    char fname[20]="test.txt";

    printf("\n\n Create a file (test.txt) and input text :\n");
    printf("-----\n");
    fptr=fopen(fname,"w");
    if(fptr==NULL)

```

```

{
    printf(" Error in opening file!");
    exit(1);
}
printf(" Input a sentence for the file : ");
fgets(str, sizeof str, stdin);
fprintf(fptr,"%s",str);
fclose(fptr);
printf("\n The file %s created successfully...!!\n\n",fname);
return 0;
}

```

```

Create a file (test.txt) and input text :
-----
Input a sentence for the file : hii this is kratik
The file test.txt created successfully...!!

```

Q=Write a program in C to read an existing file

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void main()
```

```

{

    FILE *fptr;

    char fname[20];

    char str;

    printf("\n\n Read an existing file :\n");

    printf("-----\n");

    printf(" Input the filename to be opened : ");

    scanf("%s",fname);

    fptr = fopen (fname, "r");

    if (fptr == NULL)

    {

        printf(" File does not exist or cannot be opened.\n");
    }
}

```

```

        exit(0);

    }

    printf("\n The content of the file %s is  :\n",fname);

    str = fgetc(fptr);

    while (str != EOF)

        {

            printf ("%c", str);

            str = fgetc(fptr);

        }

    fclose(fptr);

    printf("\n\n");

}

```

```

Read an existing file :
-----
Input the filename to be opened : test.txt

The content of the file test.txt is  :
hii this is kratik

```

Q=Write a program in C to Find the Number of Lines in a Text File

```

#include <stdio.h>

#define FSIZE 100

int main()
{
    FILE *fptr;
    int ctr = 0;
    char fname[FSIZE];
    char c;
    printf("\n\n Read the file and count the number of lines :\n");
    printf("-----\n");
    printf(" Input the file name to be opened : ");
    scanf("%s",fname);

    fptr = fopen(fname, "r");

```

```

if (fptr == NULL)
{
    printf("Could not open file %s", fname);
    return 0;
}
// Extract characters from file and store in character c
for (c = getc(fptr); c != EOF; c = getc(fptr))
    if (c == '\n') // Increment count if this character is newline
        ctr = ctr + 1;
fclose(fptr);
printf(" The lines in the file %s are : %d \n \n", fname, ctr-1);
return 0;
}

```

Read the file and count the number of lines :

Input the file name to be opened : test.txt
The lines in the file test.txt are : 5